EXP-NO: 9 Design a simple DATE: 29/08/25

Technology and configure with one Router, two switches and PCs using CISCO Packet Tracer.

AIM:

To design and configure an simple network Topology consisting of one router, two, Switches, and multiple PCs in CISCO Packet tracer, and Ensure communication Between PCs.

Definitions:-

1) Router:- A networking device that forwards data packets between computer networks, directing traffic on the internet.

2) Switch:- A device that connects devices within a network and uses MAC Address to forward data to the correct destination

3) PC (Personal computer): end devices in a network that users interact with to send and receive data

4). Cables: 1) Straight Through Cable

2) Crossover Cable

3) Console cable

Equipments Required

1) Cisco Packet Tracer Software
2) 1 Router (Eg: 2811)
3) 2 switches (eg: 2960)
4) 4 PCs
5) Appropriate cable (copper straight - through)

Procedure:

1) Open Cisco Packet tracer
2) Add Devices to workshop
3) connect Devices with cables
4) Configure Router Interface.
5) Configure PC IP addresses
6) Save and Test connectivity.

Result:

The PCs connected to different switches are communicate with Each other via router. The successfully Ping replies Verify the network is Proper -ly configured

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Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=3ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=6ms TTL=128
Reply from 192.168.1.10: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 3ms

C:\>
```



